Meta Learning and Meta RL

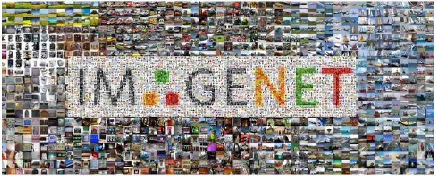
Di Wu
McGill & MILA

4th April 2019

Outline

Motivation for Meta Learning

- Deep learning models can perform well with huge amount of data



Motivation for Meta Learning

- Deep learning models struggle when
  - Training data is limited
  - Need to adapt fast to changes in the task

- Humans can generaize well with very small amount of data

Objective of Meta Learning

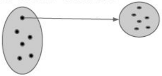
- Human can learn fast since they never learn from scratch
  - ◇ Samples
  - ◇ Models
  - ◇ Representation
  - ◇ Learning to learn
- Learning to learn/Meta Learning: Train a model on several learning tasks to solve new learning tasks with a small number of training samples.

Definition of Meta Learning

- Generalization across tasks instead of ~~data points~~. Task Level!

- What is Meta Learning / Learning to Learn?
  - Go beyond train from samples from a single distribution.

  - Distribution over tasks, so model has to "learn to learn" when a new task is presented



"... a system that improves or discovers a learning algorithm"

Hochreiter et al, '01

Oriol Vinyals, NIPS 17

What is Meta Knowledge

Anything that is determined before learning process.

- Parameter initialization
- Loss function
- Network structure
- Hyper parameters: batch size, learning rate..
- Optimization algorithm
- ...

Formalization of Meta Learning

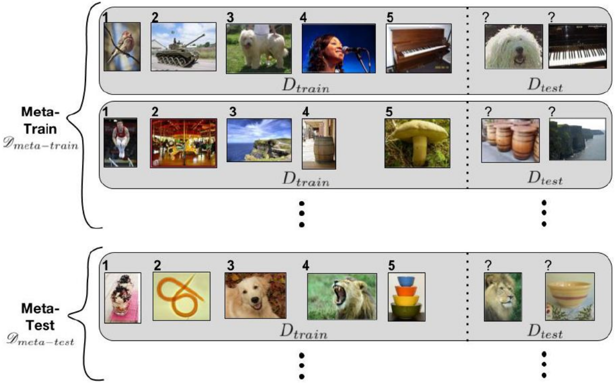
- We have a distribution of tasks $\mathcal{T}$ = $P(\mathcal{T}_i)$
- $\mathcal{T}_i$ is episodic and defined by: input x_t, output a_t, loss function $\mathcal{L}_i(x_t, a_t)$, and an episodic length H_i
- Meta learner models distribution $\pi(a_t|x_1, ..., x_t; \theta)$ and the objective is to minimize the **Meta Loss** with respect to θ:

$$\min_{\theta} \mathbb{E}_{\mathcal{T}_i \sim T}[\sum_{t=0}^{H_t} \mathcal{L}_i(x_t, a_t)]$$

Important Concepts for Meta Learning

- Meta Training: Optimizing the meta loss on sampled tasks.
- Sample set, query set.

- Meta Testing: Evaluated on unseen tasks from the same task distributions.
- Support set, test set.



Outline

1. Introduction to Meta Learning
2. Overview of Meta Learning Approach
3. Meta Reinforcement Learning
4. General Discussion: Related Problems & Research Opportunities

Early Works on Meta Learning

- Evolutionary principles in self-referential learning [Schmidhuber, 1987], applying generic programming to itself to evolve better genetic programming algorithm.
- Learning to learn [Thrun and Pratt, 1998], a discussion on learning to learn applications.
- ...

Not a new learning paradigm. Paid more attention since 2016.

Approaches for Recent Meta Learning Works

- Gradient Based: MAML [Finn et al., 2017], Reptile [Nichol and Schulman, 2018].
  - ◇ Learn a model initialization, easy to fine tune
- Metric Based: Matching Networks [Vinyals et al., 2016], Prototypical Networks [Snell et al., 2017].
  - ◇ Learn a embedding function for non-parametric method
- RNN Memory Based: MANN [Santoro et al., 2016], Learning to reinforcement learn [Wang et al., 2016] ,RL^2 [Duan et al., 2016].
  - ◇ Store meta knowledge in RNN hidden states or external memory.

Meta Supervised Learning

Classification Data Set: Omniglot and MiniImagenet.

- Omniglot (Transpose of MNIST) consists of 20 instances of 1623 characters.
- MiniImagenet: a subset of ImageNet dataset, 100 classes, each with 600 instances.

N-way, K-shot Clasification:

- K samples for each class of N classes.

Meta Learning and One-shot (Few-shot) Learning

- Firstly appeared in computer vision [Fe-Fei et al., 2003].
- One-shot (Few-shot) learning problem: Estimate models of categories from very few, one in the limit, training examples.
- Suitable for meta learning algorithms.

One Shot Learning Approaches

Approaches for One Shot Learning

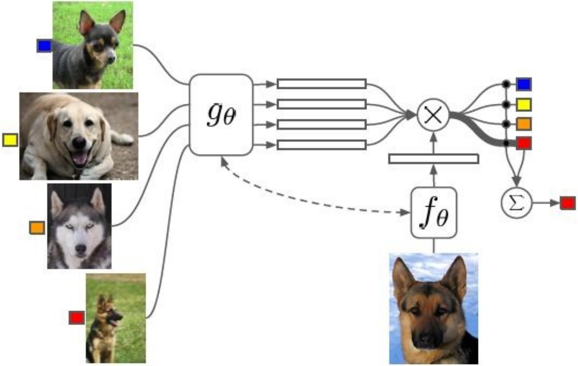
- Directly supervised learning-based approaches, e.g.,Non-parametric Methods
- Data Augmentation Methods
- Transfer Learning
- Meta Learning

Motivation for Metric-based Methods

Matching Networks [2016 NeurIPS]

- Deep learning model does not work well with small amount of data
  - ◇ Huge amount of parameters, overfitting
- Non-parametric methods: Allow novel examples be rapidly assimilated.
- Metric based methods: Combine merits of both.
  - ◇ Easy to recognize with non-parametric methods(nearest neighborhood) in the embedding space.

Model Architecture for Matching Network



Proposed Model of Matching Network

$$P(\hat{y}|\hat{x}, S) = \sum_{i=1}^{k} a(\hat{x}, x_i) y_i$$

- x_i and y_i are the inputs and corresponding labels from the support set S

- Attention mechanism/nearest neighbor: softmax over cosine distance c: $a(\hat{x}, x_i) = e^{c(f(\hat{x}), g(x_i))} / \sum_{j=1}^{K} e^{c(f(\hat{x}), g(x_j))}$

Contributions

- Matching Network model: Non parametric method with parametric neural networks
- Training philosophy: Training and testing should match (N-way, K-shot)

Motivation for Prototypical Networks

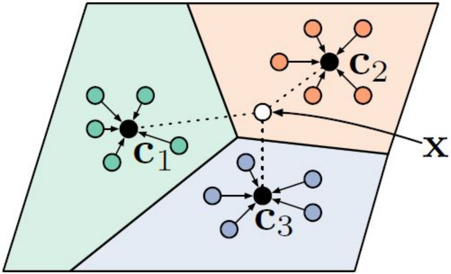
Prototypical Networks [2017 NeurIPS]

- There exists one single prototype representation for each class in the embedding space.

- Consider euclidean distance

Demo for Prototypical Networks

Prototype: Mean of all the samples of the same class.



Model of Prototypical Networks

- Class distribution computed over prototypes in embedding spaces

$$p_\phi(y = k \mid \mathbf{x}) = \frac{\exp(-d(f_\phi(\mathbf{x}), \mathbf{c}_k))}{\sum_{k'} \exp(-d(f_\phi(\mathbf{x}), \mathbf{c}_{k'}))}$$

Relationship with Matching Network

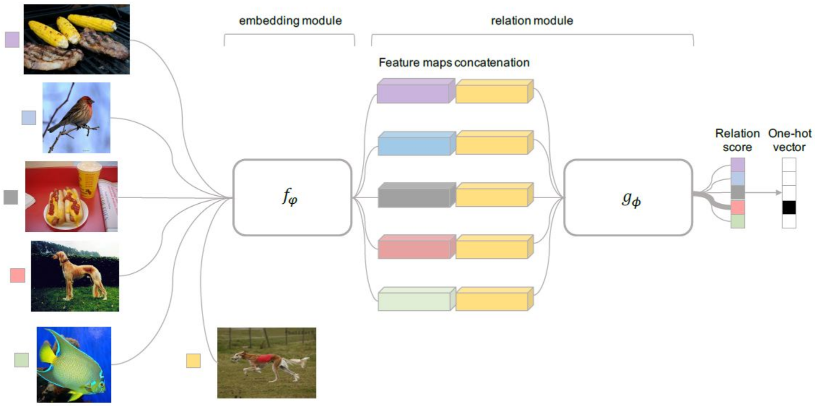
- Equivalent when k = 1
- Weighted Nearest Neighbor Method
- Euclidean distance performs better than cosine distance

Motivation for Relation Network

Relation Networks [2018 CVPR]

- Whether we should keep the metric fixed?
- Whether we can jointly learn the metric as well as the embedding?

Overview for Relation Network



Relation score

Relation score $r_{i,j}$ for the relation between querry input x_i and training sample set example x_j

$$r_{i,j} = g_\phi(\mathcal{C}(f_\varphi(x_i), f_\varphi(x_j))), \quad i = 1, 2, \ldots, C$$

Objective function

- One-shot vs K-shot: concatenating all the features of k samples.
- Treated as a regression problem.

$$\varphi, \phi \leftarrow \underset{\varphi, \phi}{\operatorname{argmin}} \sum_{i=1}^{m} \sum_{j=1}^{n} (r_{i,j} - \mathbf{1}(y_i == y_j))^2$$

Why It Works

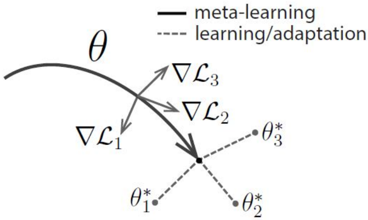
- Nonlinear in embedding space is not enough
- Jointly learning provide better performance

MAML: Model Agnostic Meta Learning

- Basic ideas: To learn a good model initialization.
- Agnostic: be applied to any models trained with gradient descent
- Learn a transferable internal representation to make the models to be easy and fast to fine-tune (maximize sensitivity).

MAML: Model Agnostic Meta Learning

Optimizing for a representation θ that can quickly adapt to new tasks.



MAML: Model Agnostic Meta Learning

Supervised learning: f_θ

- Squared error for regression
- Cross entropy for classification

Algorithm 1 Model-Agnostic Meta-Learning

Require: $p(\mathcal{T})$: distribution over tasks
Require: α, β: step size hyperparameters

```
randomly initialize θ
while not done do
    Sample batch of tasks T_i ~ p(T)
    for all T_i do
        Evaluate ∇_θ L_{T_i}(f_θ) with respect to K examples
        Compute adapted parameters with gradient descent: θ'_i = θ − α∇_θ L_{T_i}(f_θ)
    end for
    Update θ ← θ − β∇_θ Σ_{T_i~p(T)} L_{T_i}(f_{θ'_i})
end while
```

$$\theta'_i = \theta - \alpha \nabla_\theta \mathcal{L}_{\mathcal{T}_i}(f_\theta)$$

$$\theta \leftarrow \theta - \beta \nabla_\theta \sum_{\mathcal{T}_i \sim p(\mathcal{T})} \mathcal{L}_{\mathcal{T}_i}(f_{\theta'_i})$$

MAML: Model Agnostic Meta Learning

Gradient over Meta Loss

$$\min_{\theta} \sum_{\mathcal{T}_i \sim p(\mathcal{T})} \mathcal{L}_{\mathcal{T}_i}(f_{\theta'_i}) = \sum_{\mathcal{T}_i \sim p(\mathcal{T})} \mathcal{L}_{\mathcal{T}_i}(f_{\theta - \alpha \nabla_\theta \mathcal{L}_{\mathcal{T}_i}(f_\theta)})$$

MAML: Model Agnostic Meta Learning

MAML for supervised learning

Algorithm 2 MAML for Few-Shot Supervised Learning

Require: $p(\mathcal{T})$: distribution over tasks
Require: α, β: step size hyperparameters
1: randomly initialize θ
2: **while** not done **do**
3: Sample batch of tasks $\mathcal{T}_i \sim p(\mathcal{T})$
4: **for all** $\mathcal{T}_i$ **do**
5: Sample K datapoints $\mathcal{D} = \{\mathbf{x}^{(j)}, \mathbf{y}^{(j)}\}$ from $\mathcal{T}_i$
6: Evaluate $\nabla_\theta \mathcal{L}_{\mathcal{T}_i}(f_\theta)$ using $\mathcal{D}$ and $\mathcal{L}_{\mathcal{T}_i}$ in Equation (2) or (3)
7: Compute adapted parameters with gradient descent: $\theta_i' = \theta - \alpha \nabla_\theta \mathcal{L}_{\mathcal{T}_i}(f_\theta)$
8: Sample datapoints $\mathcal{D}_i' = \{\mathbf{x}^{(j)}, \mathbf{y}^{(j)}\}$ from $\mathcal{T}_i$ for the meta-update
9: **end for**
10: Update $\theta \leftarrow \theta - \beta \nabla_\theta \sum_{\mathcal{T}_i \sim p(\mathcal{T})} \mathcal{L}_{\mathcal{T}_i}(f_{\theta_i'})$ using each $\mathcal{D}_i'$ and $\mathcal{L}_{\mathcal{T}_i}$ in Equation 2 or 3
11: **end while**

Experimental Results on Omniglot

Results on Omniglot

Model	Fine Tune	5-way Acc.		20-way Acc.	
		1-shot	5-shot	1-shot	5-shot
MANN [32]	N	82.8%	94.9%	-	-
CONVOLUTIONAL SIAMESE NETS [20]	N	96.7%	98.4%	88.0%	96.5%
CONVOLUTIONAL SIAMESE NETS [20]	Y	97.3%	98.4%	88.1%	97.0%
MATCHING NETS [39]	N	98.1%	98.9%	93.8%	98.5%
MATCHING NETS [39]	Y	97.9%	98.7%	93.5%	98.7%
SIAMESE NETS WITH MEMORY [18]	N	98.4%	99.6%	95.0%	98.6%
NEURAL STATISTICIAN [8]	N	98.1%	99.5%	93.2%	98.1%
META NETS [27]	N	99.0%	-	97.0%	-
PROTOTYPICAL NETS [36]	N	98.8%	99.7%	96.0%	98.9%
MAML [10]	Y	$98.7 \pm 0.4\%$	**$99.9 \pm 0.1\%$**	$95.8 \pm 0.3\%$	$98.9 \pm 0.2\%$
RELATION NET	N	**$99.6 \pm 0.2\%$**	**$99.8 \pm 0.1\%$**	**$97.6 \pm 0.2\%$**	**$99.1 \pm 0.1\%$**

Experimental Results on MiniImagenet

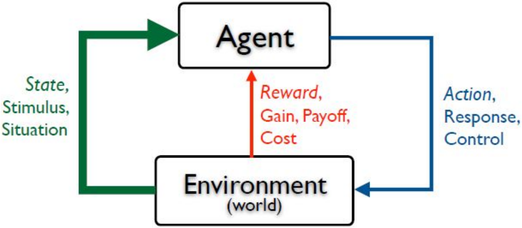
Results on MiniImagenet

Model	FT	5-way Acc.	
		1-shot	5-shot
MATCHING NETS [39]	N	$43.56 \pm 0.84\%$	$55.31 \pm 0.73\%$
META NETS [27]	N	$49.21 \pm 0.96\%$	-
META-LEARN LSTM [29]	N	$43.44 \pm 0.77\%$	$60.60 \pm 0.71\%$
MAML [10]	Y	$48.70 \pm 1.84\%$	$63.11 \pm 0.92\%$
PROTOTYPICAL NETS [36]	N	$49.42 \pm 0.78\%$	$\mathbf{68.20 \pm 0.66\%}$
RELATION NET	N	$\mathbf{50.44 \pm 0.82\%}$	$65.32 \pm 0.70\%$

Outline

Reinforcement Learning Framework

Objective:
Enable an agent to quickly learn a policy for a new task with a small amount of experiences/interactions.



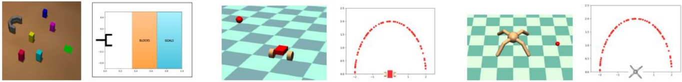
Reinforcement Learning Tasks

Differences compared with meta supervised learning

- Loss function
- How data is gathered and presented

Changes for RL tasks

- Achieve a new goal in the same environment.
- Achieve the same goal in a different environment.



(a) Robotic Manipulation (b) Wheeled Locomotion (c) Legged Locomotion

Meta Knowledge for Reinforcement Learning

RL meta knowledge

- Reward function: γ, λ
- Exploration strategy
- Model initialization...

Suitable approaches

- Gradient based methods
- RNN memory based methods

MAML for Reinforcement Learning

Loss function: Negative of the reward function R

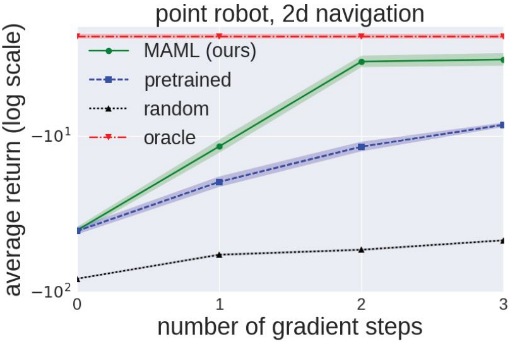
Algorithm 3 MAML for Reinforcement Learning

Require: $p(\mathcal{T})$: distribution over tasks
Require: α, β: step size hyperparameters
1: randomly initialize θ
2: **while** not done **do**
3: Sample batch of tasks $\mathcal{T}_i \sim p(\mathcal{T})$
4: **for all** $\mathcal{T}_i$ **do**
5: Sample K trajectories $\mathcal{D} = \{(\mathbf{x}_1, \mathbf{a}_1, ...\mathbf{x}_H)\}$ using f_θ in $\mathcal{T}_i$
6: Evaluate $\nabla_\theta \mathcal{L}_{\mathcal{T}_i}(f_\theta)$ using $\mathcal{D}$ and $\mathcal{L}_{\mathcal{T}_i}$ in Equation 4
7: Compute adapted parameters with gradient descent: $\theta_i' = \theta - \alpha \nabla_\theta \mathcal{L}_{\mathcal{T}_i}(f_\theta)$
8: Sample trajectories $\mathcal{D}_i' = \{(\mathbf{x}_1, \mathbf{a}_1, ...\mathbf{x}_H)\}$ using $f_{\theta_i'}$ in $\mathcal{T}_i$
9: **end for**
10: Update $\theta \leftarrow \theta - \beta \nabla_\theta \sum_{\mathcal{T}_i \sim p(\mathcal{T})} \mathcal{L}_{\mathcal{T}_i}(f_{\theta_i'})$ using each $\mathcal{D}_i'$ and $\mathcal{L}_{\mathcal{T}_i}$ in Equation 4
11: **end while**

MAML for Reinforcement Learning

Results for 2d navigation tasks.

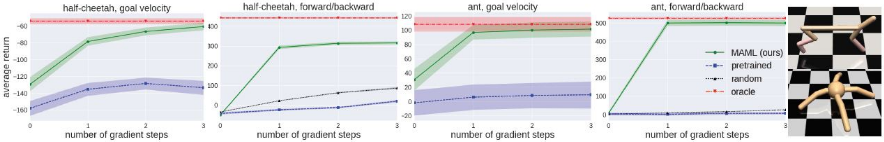
- Goal position is randomly chosen within a unit square.



Experimental Results

Results for half-cheetah and ant locomotion tasks tasks.

- The goal of velocity or direction is chosen randomly for each task.



Representative Works

- **RNN Based**
- *Learning to reinforcement learn* [Wang et al., 2016]
- *RL2: Fast Reinforcement Learning via Slow Reinforcement Learning* [Duan et al., 2016]
- *A simple neural attentive meta-learner* [Mishra et al., 2017]
- **Initialization**
- *Learn the model initialization for fast adaptation* [Finn et al., 2017].
- **Hierarchial RL**
- *Meta learning shared hierarchies* [Frans et al., 2017]

Representative Works

- **Exploration**
- *Meta-reinforcement learning of structured exploration strategies* [Gupta et al., 2018]
- *Some considerations on learning to explore via meta-reinforcement learning* [Stadie et al., 2018]
- *Learning to explore with meta-policy gradient* [Xu et al., 2018a]
- **Reward Function**
- *Meta-gradient reinforcement learning* [Xu et al., 2018b]
- *Evolved policy gradients* [Houthooft et al., 2018]
- **Credit Assignment**
- *ProMP: Proximal Meta-Policy Search* [Rothfuss et al., 2018]

Representative Works

- **Model Based RL**
- *Learning to adapt: Meta-learning for model-based control* [Clavera et al., 2018]
- *Learning to Adapt in Dynamic, Real-World Environments through Meta-Reinforcement Learning* [Nagabandi et al., 2018a]
- *Deep Online Learning via Meta-Learning: Continual Adaptation for Model-Based RL* [Nagabandi et al., 2018b]
- **Inverse RL**
- *Learning a prior over intent via meta-inverse reinforcement learning* [Xu et al., 2018a]

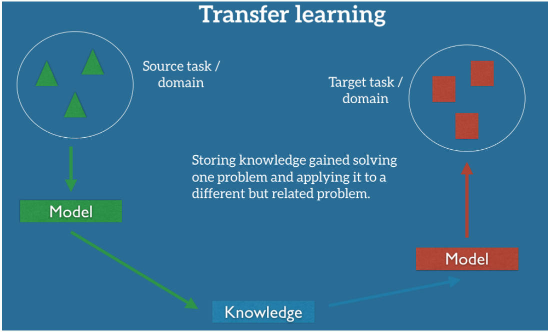
Outline

Related Learning Paradigms

- **Knowledge Reuse via Prior Learning**
  - ◇ *Transfer Learning*
  - ◇ *Multi-task Learning*
  - ◇ *Lifelong & Continual Learning*

Transfer Learning

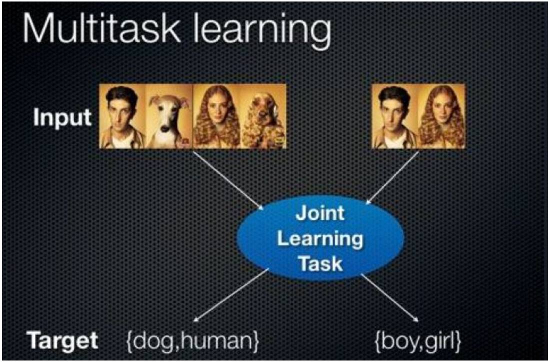
- Transfer learning: Leverage the information from the source domain(s) to help learning in the **target domain**.
- Meta Learning: A **higher level** knowledge transfer.
- Focus on **learning fast** (as well as accumulated performance).



Multi-task Learning

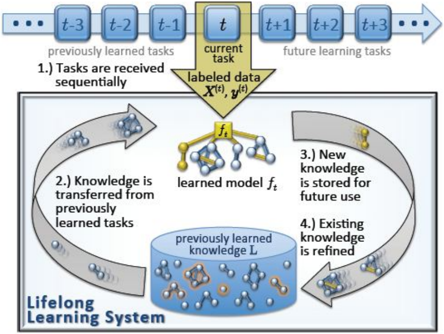
- Multi-task Learning: Optimize learning/performance across all tasks through shared knowledge.
- Multi-task Learning highly dependent on the model.
- Meta Learning: A kind of **historical** multi-task.

Example for image classification.



Lifelong Learning and Continual Learning

- Lifelong Learning: Leverage the relevant knowledge gained in the past N-1 tasks to help learning for the Nth task.
- Continual Learning: Similar concept and more focused on catastrophic forgetting for neural networks.
- Online version and batch version.



Potential Research Directions

Algorithms

- *Integrate different Meta Learning Approaches*
- *Limitations of Current Meta Learning Approaches*
- *Meta RL: Exploration, Learning Rate, HRL, Model Based*
- *Generative Models*
- *Continual Learning*

Potential Research Opportunities

Applications Go beyond Omniglot and MiniImagenet

- *Low Resource NLP: Few-shot QA*
- *Few-shot Speech to Text*
- *Few-shot Generation: Missing Data*
- *Fast Adaptation in Real-world Control Problems*

Q&A


thank you
danke
謝謝
ngiyabonga
teşekkür ederim
gracias
merci
obrigado
bedankt
спасибо
Баярлалаа
рахмат
dziękuje
sukriya
kop khun krap
go raibh maith agat
mochchakkeram
tapadh leat
dank je
grazie
arigatô
dakujem
terima kasih
감사합니다
ありがとう
мерси
ευχαριστώ
takk
taiku
sagolun
hvala
maururu
kiitos
dankie
dhanyavad
nanni
nandri
enkosi
faafetai lava
vinaka
спасибі
blagodaram
mersi
barka
welalin
tack
spas
misaotra
matondo
paldies
grazzi
mahalo
хвала
asante
manana
obrigada
murakoze
tenki
djiere dieuf
дякую
mamnun
chokrane
trugarez
shukriya
mercê
dhanyavadagalu
diolch
rahmet
tanemirt
xièxie
sulpáy
grácies
gratias ago
chnorakaloutioun
dankon
akun
köszönöm
gràcie
bayarlalaa
sobodi
dėkuji
mèsi
didi madloba
najis tuke
kam sah hamnida
rahmat
তোমাকে ধন্যবাদ

Reference I

Clavera, I., Nagabandi, A., Fearing, R. S., Abbeel, P., Levine, S., and Finn, C. (2018).
Learning to adapt: Meta-learning for model-based control.
arXiv preprint arXiv:1803.11347.

Duan, Y., Schulman, J., Chen, X., Bartlett, P. L., Sutskever, I., and Abbeel, P. (2016).
Rl2: Fast reinforcement learning via slow reinforcement learning.
arXiv preprint arXiv:1611.02779.

Fe-Fei, L. et al. (2003).
A bayesian approach to unsupervised one-shot learning of object categories.
In Proceedings Ninth IEEE International Conference on Computer Vision, pages 1134–1141. IEEE.

Reference II

- Finn, C., Abbeel, P., and Levine, S. (2017).
  Model-agnostic meta-learning for fast adaptation of deep networks.
  In Proceedings of the 34th International Conference on Machine Learning-Volume 70, pages 1126–1135. JMLR. org.

- Frans, K., Ho, J., Chen, X., Abbeel, P., and Schulman, J. (2017).
  Meta learning shared hierarchies.
  arXiv preprint arXiv:1710.09767.

- Gupta, A., Mendonca, R., Liu, Y., Abbeel, P., and Levine, S. (2018).
  Meta-reinforcement learning of structured exploration strategies.
  In Advances in Neural Information Processing Systems, pages 5307–5316.

Reference III

Houthooft, R., Chen, Y., Isola, P., Stadie, B., Wolski, F., Ho, O. J., and Abbeel, P. (2018).
Evolved policy gradients.
In Advances in Neural Information Processing Systems, pages 5405–5414.

Mishra, N., Rohaninejad, M., Chen, X., and Abbeel, P. (2017).
A simple neural attentive meta-learner.
arXiv preprint arXiv:1707.03141.

Nagabandi, A., Clavera, I., Liu, S., Fearing, R. S., Abbeel, P., Levine, S., and Finn, C. (2018a).
Learning to adapt in dynamic, real-world environments through meta-reinforcement learning.

Reference IV

Nagabandi, A., Finn, C., and Levine, S. (2018b).
Deep online learning via meta-learning: Continual adaptation for model-based rl.
arXiv preprint arXiv:1812.07671.

Nichol, A. and Schulman, J. (2018).
Reptile: a scalable metalearning algorithm.
arXiv preprint arXiv:1803.02999, 2.

Rothfuss, J., Lee, D., Clavera, I., Asfour, T., and Abbeel, P. (2018).
Promp: Proximal meta-policy search.
arXiv preprint arXiv:1810.06784.

Reference V

- Santoro, A., Bartunov, S., Botvinick, M., Wierstra, D., and Lillicrap, T. (2016).
  One-shot learning with memory-augmented neural networks.
  arXiv preprint arXiv:1605.06065.

- Schmidhuber, J. (1987).
  Evolutionary principles in self-referential learning.
  On learning how to learn: The meta-meta-... hook.) Diploma thesis, Institut f. Informatik, Tech. Univ. Munich.

- Snell, J., Swersky, K., and Zemel, R. (2017).
  Prototypical networks for few-shot learning.
  In Advances in Neural Information Processing Systems, pages 4077–4087.

Reference VI

Stadie, B. C., Yang, G., Houthooft, R., Chen, X., Duan, Y., Wu, Y., Abbeel, P., and Sutskever, I. (2018).
Some considerations on learning to explore via meta-reinforcement learning.
arXiv preprint arXiv:1803.01118.

Thrun, S. and Pratt, L. (1998).
Learning to learn.

Vinyals, O., Blundell, C., Lillicrap, T., Wierstra, D., et al. (2016).
Matching networks for one shot learning.
In Advances in neural information processing systems, pages 3630–3638.

Reference VII

- Wang, J. X., Kurth-Nelson, Z., Tirumala, D., Soyer, H., Leibo, J. Z., Munos, R., Blundell, C., Kumaran, D., and Botvinick, M. (2016).
  Learning to reinforcement learn.
  arXiv preprint arXiv:1611.05763.

- Xu, T., Liu, Q., Zhao, L., Xu, W., and Peng, J. (2018a).
  Learning to explore with meta-policy gradient.
  arXiv preprint arXiv:1803.05044.

- Xu, Z., van Hasselt, H. P., and Silver, D. (2018b).
  Meta-gradient reinforcement learning.
  In Advances in Neural Information Processing Systems, pages 2402–2413.